

Assignment Na-03

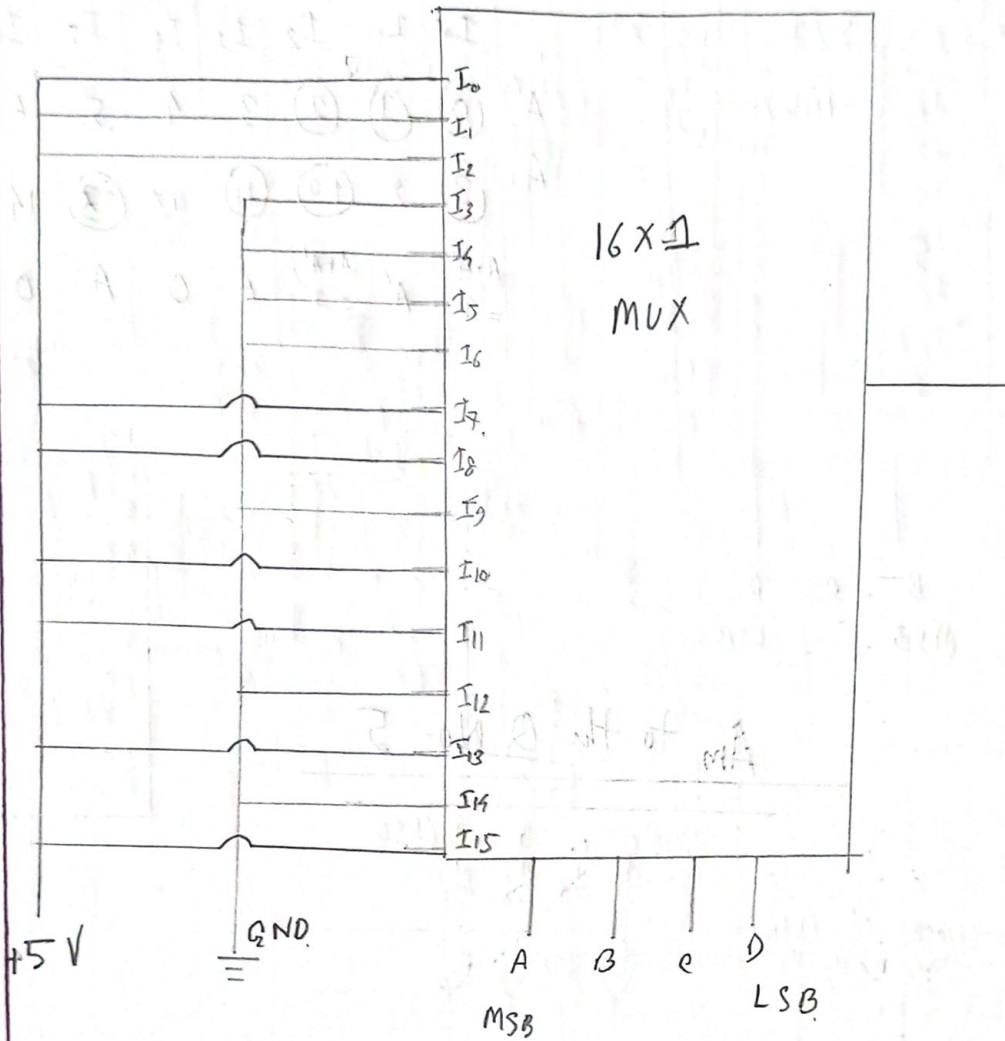
Name: Tahsin Raihan Robbani

ID : 22201344

Section: 1a

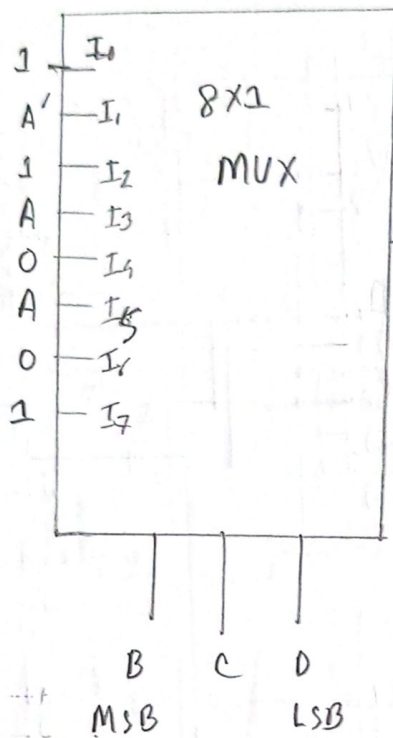
Ans to the Q No-2

(a) $F(A, B, C, D) = \sum (0, 1, 2, 7, 8, 10, 11, 13, 15)$



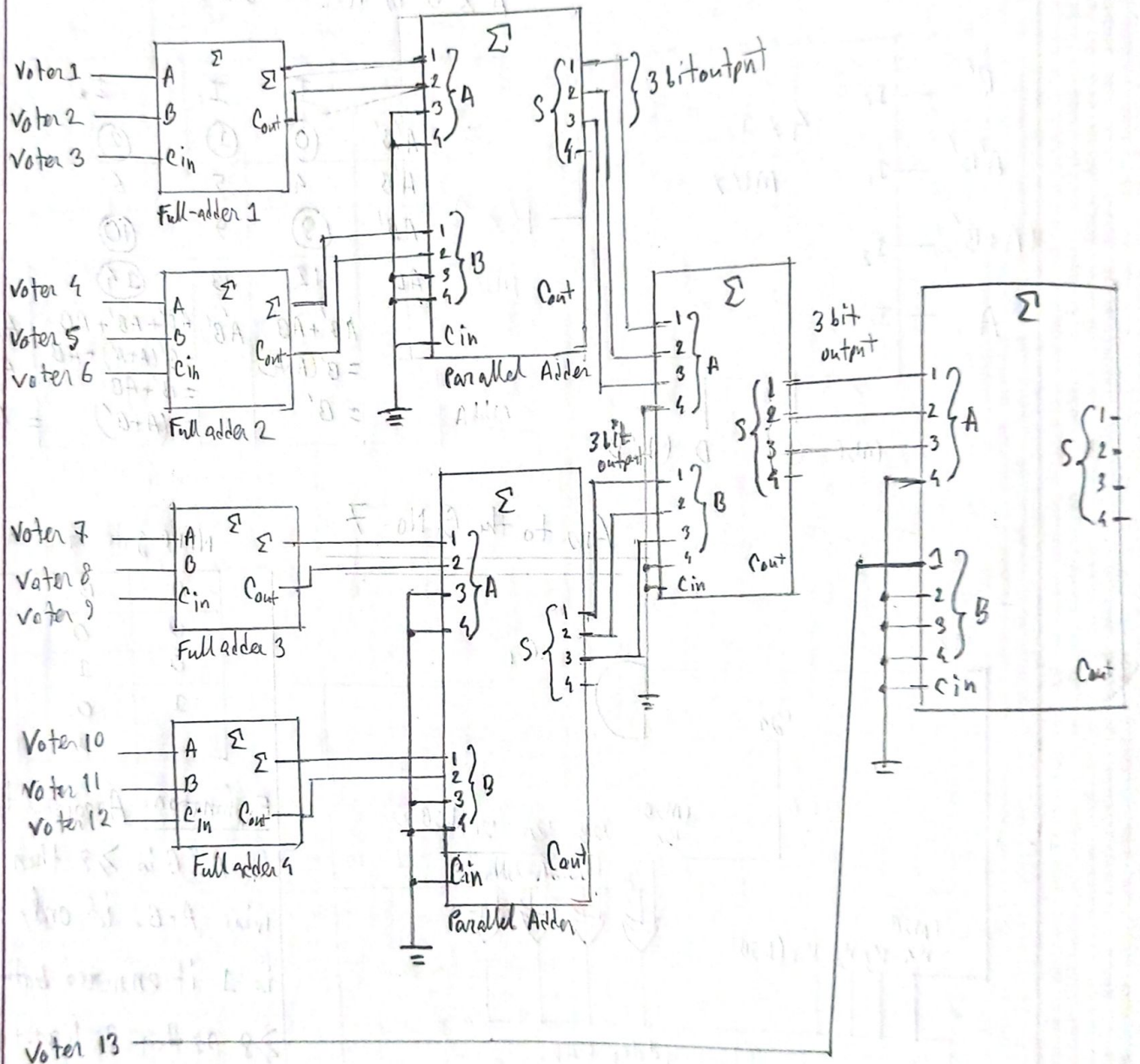
(b) $F(A, B, C, D) = \Sigma(0, 1, 2, 7, 8, 9, 11, 13, 15)$

as selector
A is not used ~~in minterms~~

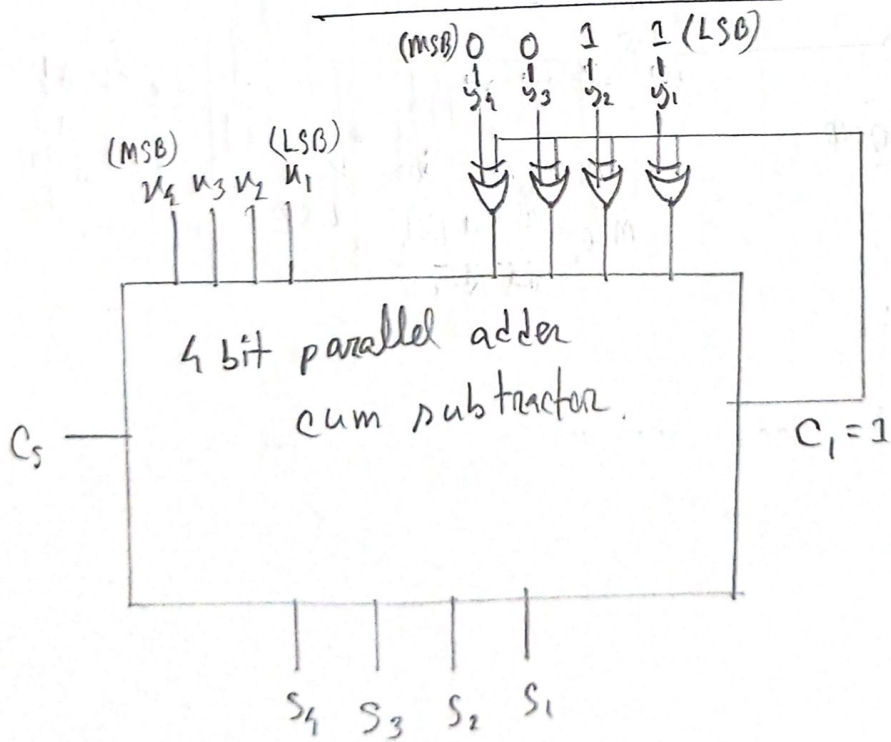


	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
$(A+A') = 1$	A'	$(A+A') = 1$	A	0	A	0	$(A+A') = 1$	

Ans to the Q No-4



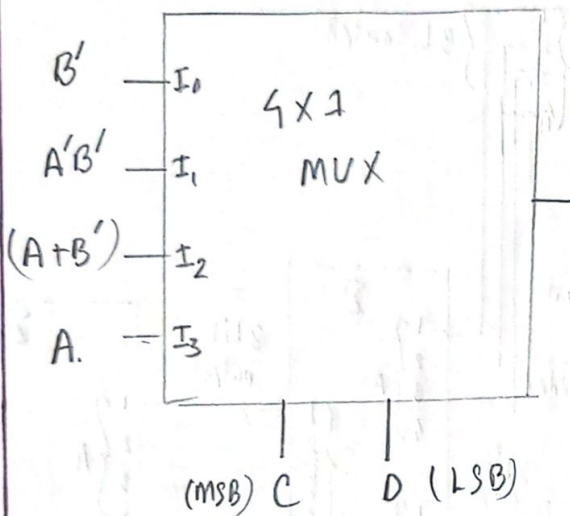
Ans to the Q No- 5



Ans to the Q No-6

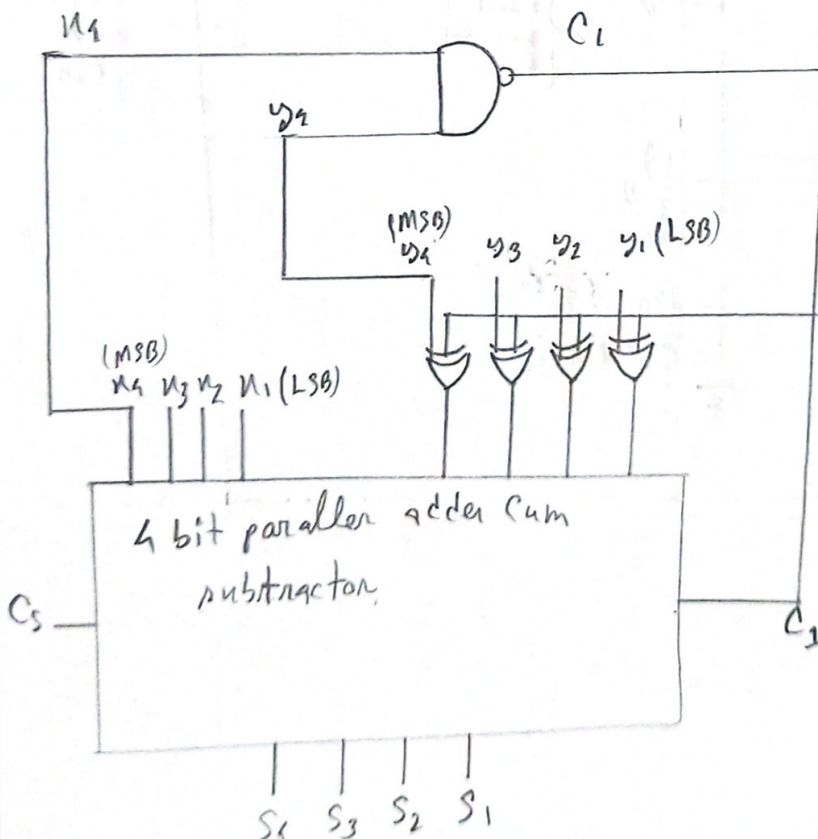
$$F(A, B, C, D) = \sum (0, 1, 2, 8, 10, 11, 13, 15)$$

A & B is not used as selector.



	I_0	I_1	I_2	I_3
$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15
$A'B' + AB'$ $= B'(A+A')$ $= B'$		$A'B'$	$A'B' + AB' + AB$ $= B'(A+A') + AB$ $= B' + AB$ $= (A+B')$	$AB' + AB$ $= A(B+B')$ $= A$

Ans to the Q No-7



NAND gate Truth table

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

Explanation: According to question if $A \& B$ is ≥ 8 then $A+B$ other wise $A-B$. if only u_4 & u_3 is 1 it ensures both value is ≥ 8 so then C_1 has to be zero for addition. Only with NAND gate we can fulfill this condition. Similarly one value is < 8 our C_1 will give 1 which means $A-B$.